RAZORPOWER AERONAUTICS SPACE ADMINISTRATION
R A S A

U.S.S. KEEPER OF TOMORROW

CARGO BAY 12
PLC OPERATIONS MANUAL

STANDARD OPERATING PROCEDURE
CARGO SHIPPING AND RECEIVING OPERATORS

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DOCUMENT CONTROL	
Document	Cargo Bay 12 PLC Operations Manual
Vessel	U.S.S. Keeper of Tomorrow
Operating authority	RASA (Razorpower Aeronautics Space Administration)
Applies to	Cargo Bay 12, PLC runtime 172.30.0.100
Audience	Cargo Shipping and Receiving Operators
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1. PURPOSE

This manual tells Cargo Shipping and Receiving Operators how the programmable logic controller (PLC) in Cargo Bay 12 is wired, what its signals mean, what it does automatically, and how to respond to what it reports. It combines the bay's I/O tag map, its ladder logic reference, and its network layout into one document.

2. SCOPE

This procedure applies to Cargo Bay 12 aboard the U.S.S. Keeper of Tomorrow. The bay has four doors: one bay entry door (door_0) and three container doors (door_1, door_2, door_3). The three container doors face the exterior of the spacecraft and lead to the containers carried on board. Each cargo bay is equipped with a set of sensors, a programmable logic controller, and an uplink.

It covers the PLC's documented inputs and outputs, the four programmed functions it runs, and the network hosts it communicates with. It does not replace RASA flight, safety, or cargo-handling regulations.

3. SYSTEM OVERVIEW

Cargo Bay 12 is controlled by a single PLC runtime at 172.30.0.100 (port 502). The PLC continuously repeats the same cycle:

1) **READ** the input signals from the doors, container sensors, and shutdown input.
2) **SOLVE** the ladder logic rungs using those input values.
3) **WRITE** the results to the output tags that drive the camera, alarms, and lights, and send commands back to the doors and shutdown device.

The bay's field devices are: one camera, three container alarms, four door devices (the bay entry door and three container doors), three cargo lights, one pressure alarm, three container sensors, and one shutdown input. Section 4 shows how they connect to the PLC and Section 5 lists every signal.

3.1 CONTAINER ASSIGNMENTS

Each container has its own door, sensor, and alarm. They are numbered so that the three container doors line up with the three containers:

Container	Door (status / command)	Sensor	Alarm
Container 0	door_1	sensor_0	container_alarm_0
Container 1	door_2	sensor_1	container_alarm_1
Container 2	door_3	sensor_2	container_alarm_2

Note that the door number is one higher than the container number, because door_0 is the bay entry door and belongs to no container. A container sensor reads TRUE when its container is missing, and its alarm then turns on.

4. NETWORK TOPOLOGY

Figure 1 shows every host on the Cargo Bay 12 network. Each field device is shown by its device name and address; the PLC is at the top. Devices on the left supply signals the PLC reads, and devices on the right carry signals the PLC writes.

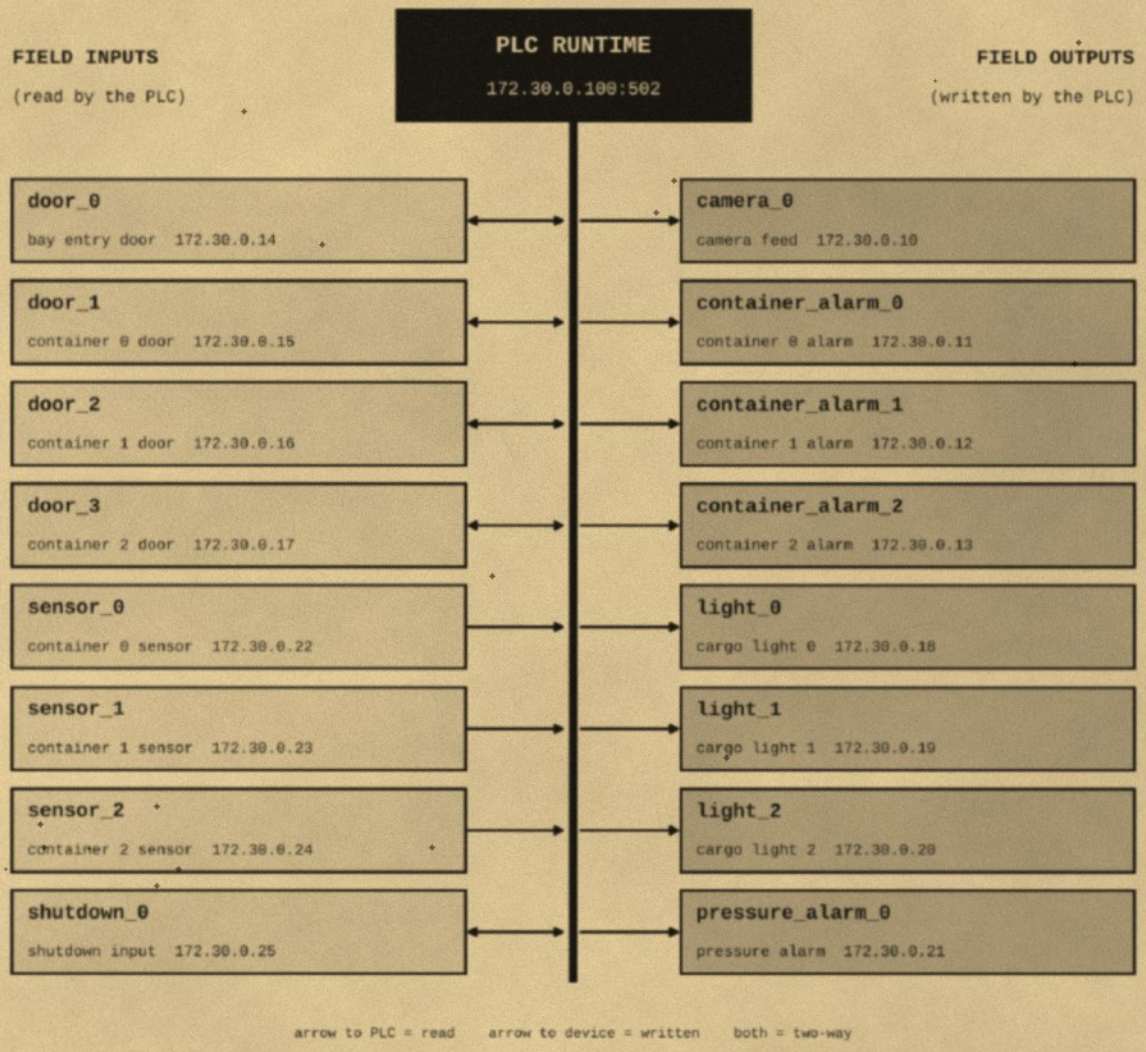


FIG. 1 CARGO BAY 12 NETWORK: PLC AND FIELD DEVICES

Note: each door device and the shutdown device has both a status input and a command output in the tag map, so it communicates in both directions. Section 5.1 explains how the command and status signals relate.

4.1 HOSTS AND ADDRESSES

Each device name matches the prefix of its tag names, so the tags in Section 5 can be traced to a device and address below.

Device	IP address	Tags	Role
PLC runtime	172.30.0.100:502	all tags	Controller
camera_0	172.30.0.10	camera_0_write	Camera feed
container_alarm_0	172.30.0.11	container_alarm_0_write	Container 0 alarm
container_alarm_1	172.30.0.12	container_alarm_1_write	Container 1 alarm
container_alarm_2	172.30.0.13	container_alarm_2_write	Container 2 alarm
door_0	172.30.0.14	door_0, door_0_command	Bay entry door
door_1	172.30.0.15	door_1, door_1_command	Container 0 door
door_2	172.30.0.16	door_2, door_2_command	Container 1 door
door_3	172.30.0.17	door_3, door_3_command	Container 2 door
light_0	172.30.0.18	light_0_write	Cargo light 0
light_1	172.30.0.19	light_1_write	Cargo light 1
light_2	172.30.0.20	light_2_write	Cargo light 2
pressure_alarm_0	172.30.0.21	pressure_alarm_0_write	Pressure alarm
sensor_0	172.30.0.22	sensor_0	Container 0 sensor
sensor_1	172.30.0.23	sensor_1	Container 1 sensor
sensor_2	172.30.0.24	sensor_2	Container 2 sensor
shutdown_0	172.30.0.25	shutdown_0, shutdown_0_command	Shutdown input

5. PLC TAG MAP

The table below lists every PLC address with its tag name and signal direction. Inputs (%IX) are read from the field; outputs (%QX) are written by the PLC to the field.

Address	Tag name	Signal	Function
%IX0.0	door_0	**INPUT**	Bay entry door status
%IX0.1	door_1	**INPUT**	Door status, container 0
%IX0.2	door_2	**INPUT**	Door status, container 1
%IX0.3	door_3	**INPUT**	Door status, container 2
%IX0.4	sensor_0	**INPUT**	Sensor: container 0 missing
%IX0.5	sensor_1	**INPUT**	Sensor: container 1 missing
%IX0.6	sensor_2	**INPUT**	Sensor: container 2 missing
%IX0.7	shutdown_0	**INPUT**	Shutdown signal
%QX0.0	camera_0_write	**OUTPUT**	Camera enable
%QX0.1	container_alarm_0_write	**OUTPUT**	Alarm: container 0 missing
%QX0.2	container_alarm_1_write	**OUTPUT**	Alarm: container 1 missing
%QX0.3	container_alarm_2_write	**OUTPUT**	Alarm: container 2 missing
%QX0.4	door_0_command	**OUTPUT**	Command to door_0
%QX0.5	door_1_command	**OUTPUT**	Command to door_1
%QX0.6	door_2_command	**OUTPUT**	Command to door_2
%QX0.7	door_3_command	**OUTPUT**	Command to door_3
%QX1.0	light_0_write	**OUTPUT**	Cargo light 0
%QX1.1	light_1_write	**OUTPUT**	Cargo light 1
%QX1.2	light_2_write	**OUTPUT**	Cargo light 2
%QX1.3	pressure_alarm_0_write	**OUTPUT**	Alarm: container door open
%QX1.4	shutdown_0_command	**OUTPUT**	Command to shutdown_0

ADDRESS FORMAT. %IX0.4 means input bit 4 in byte 0; %QX1.3 means output bit 3 in byte 1. Eight bits make one byte, so the first byte of inputs is %IX0.0 to %IX0.7.

5.1 SIGNAL CONVENTIONS

Tag(s)	TRUE means	FALSE means
door_0 to door_3 (inputs)	Door closed	Door open
door_0_command to door_3_command	Open the door	Close the door
sensor_0 to sensor_2 (inputs)	Container missing	Container present
container_alarm_n_write	Alarm on: container n missing	Alarm off
pressure_alarm_0_write	Alarm on: a container door is open	Alarm off: all container doors closed
shutdown_0 (input)	Shutdown active	Shutdown inactive (normal)
shutdown_0_command	Request shutdown	Request no shutdown
*_write outputs	Device ON / alarm sounding	Device OFF / alarm silent

- **_write TAGS** are written by the PLC only. Their values are driven by the inputs through the ladder logic in Section 7, and the field device simply acts on the value it receives (camera on, alarm on, light on).
- **_command TAGS** are written by the PLC back to the field device that owns the matching input tag. For example, door_0_command is sent to the door_0 device, which then updates door_0.
- **DOOR DEVICES INVERT THE COMMAND.** Doors are normally closed. A door device sets its status to the opposite of the command it receives, so a command of TRUE gives a status of FALSE (open), and a command of FALSE gives a status of TRUE (closed).
- **THE SHUTDOWN DEVICE DOES NOT INVERT.** shutdown_0 is normally FALSE (inactive) and the device sets shutdown_0 to the command value as is, so a command of TRUE gives a status of TRUE (shutdown active).

6. LADDER LOGIC REFERENCE

The PLC program is written in ladder logic. A rung is read left to right: conditions on the left decide whether power “flows” to the output on the right.

6.1 NORMALLY OPEN (NO) CONTACT

A normally open contact conducts when its tag is TRUE. While the tag is FALSE, the contact is open and blocks the rung. Use it to mean “do this when the signal is present.”

6.2 NORMALLY CLOSED (NC) CONTACT

A normally closed contact conducts when its tag is FALSE. While the tag is TRUE, the contact is open and blocks the rung. Use it to mean “do this when the signal is absent.”

Tag value	NO contact	NC contact
TRUE	Conducts	Blocks
FALSE	Blocks	Conducts

6.3 SERIES AND PARALLEL LOGIC

- **SERIES:** contacts in series are AND conditions. All must conduct for the rung to be true.
- **PARALLEL:** parallel branches are OR conditions. Any one branch conducting makes the rung true.

6.4 OTE COIL (OUTPUT ENERGIZE)

An OTE coil is the output instruction at the right end of a rung. It writes the rung result to its output tag every scan: when the rung conducts, the tag is set TRUE; when it does not, the tag is set FALSE. The coil does not latch, so it follows the rung and the output turns off as soon as the conditions are no longer met. The tag it writes is mapped to a physical device through the tag map in Section 5.

7. PROGRAMMED FUNCTIONS OF THE CARGO BAY 12 PLC

The PLC in Cargo Bay 12 is equipped to perform the four functions below. Each runs independently on every scan.

Function	Behavior	Rung logic	Output tag(s)
Camera program	Keeps the camera feed enabled while the shutdown signal is inactive.	NC shutdown_0	camera_0_write
Pressure alarm	Turns on the alarm when any of the three container door signals makes its normally-closed contact conduct, i.e., when that signal is false (door open). It reports an open door only; it does not check for a missing container.	NC door_1 OR NC door_2 OR NC door_3	pressure_alarm_0_write
Container alarms	Each container sensor (TRUE = container missing) independently turns on its corresponding alarm.	NO sensor_n	container_alarm_n_write

Function	Behavior	Rung logic	Output tag(s)
Cargo lights	Keeps all three lights on while the shutdown signal is inactive.	NC shutdown_0	light_0_write, light_1_write, light_2_write

7.1 CAMERA PROGRAM

An NC contact on shutdown_0 (%IX0.7) feeds an OTE coil on camera_0_write (%QX0.0). While shutdown_0 is FALSE (inactive) the contact conducts and the camera feed is on. When shutdown_0 goes TRUE, the camera feed is switched off.

7.2 PRESSURE ALARM

Three NC contacts, one each on door_1, door_2 and door_3 (the container doors for containers 0, 1 and 2), sit in parallel (OR) and feed an OTE coil on pressure_alarm_0_write (%QX1.3). If any one of these door signals is FALSE, meaning that container door is open, its NC contact conducts and the pressure alarm turns on. The alarm stays off only while all three container doors are closed (TRUE). The bay entry door (door_0) is not part of this rung.

The pressure alarm therefore means one thing: at least one container door is open. By itself it does not mean the bay is exposed to the vacuum of space. A container door opens onto its container, so the bay is exposed only when that container is also missing. The PLC does not check for this combination. Section 7.6 explains how to read the two alarms together.

7.3 CONTAINER ALARMS

Each container sensor drives its own alarm through a separate rung, so one container never affects another. A container alarm means its sensor reads TRUE, which is how the system reports that the container is missing. A container alarm on its own is acceptable as long as the door of that container stays closed, because the closed door keeps the bay sealed. The container alarms do not feed the pressure alarm; the pressure alarm responds only to open container doors.

Sensor input	Contact	Alarm output	Container / door
sensor_0 (%IX0.4)	NO	container_alarm_0_write (%QX0.1)	Container 0 / door_1
sensor_1 (%IX0.5)	NO	container_alarm_1_write (%QX0.2)	Container 1 / door_2
sensor_2 (%IX0.6)	NO	container_alarm_2_write (%QX0.3)	Container 2 / door_3

7.4 CARGO LIGHTS

An NC contact on shutdown_0 (%IX0.7) feeds OTE coils on light_0_write (%QX1.0), light_1_write (%QX1.1) and light_2_write (%QX1.2). All three lights stay on while shutdown_0 is FALSE, and all three go off when it is TRUE.

7.5 EFFECT OF THE SHUTDOWN SIGNAL

> **[NOTE]** When shutdown_0 becomes TRUE, the camera feed and all three cargo lights turn off together. The alarms are not tied to the shutdown signal and keep following their own conditions.

7.6 READING THE ALARMS TOGETHER: MISSING CONTAINER AND OPEN DOOR

The dangerous condition is a missing container together with an open door to that same container. Each alarm is only half of the picture. Always check both for the same container: alarm 0 with door_1, alarm 1 with door_2, alarm 2 with door_3. A missing container 0 with door_2 open, for example, is not the dangerous combination.

Container n	Its door	Meaning
Present	Closed	Normal. No alarm from this container.
Present	Open	Pressure alarm ON. The door opens onto the container, so the bay is not exposed to the vacuum.
Missing	Closed	Container alarm ON only. Acceptable: the closed door keeps the bay sealed.
Missing	**Open**	**EMERGENCY. Container alarm AND pressure alarm are both ON: the bay is exposed to the vacuum of space.**

> **[WARNING]** A missing container AND an open door to that container at the same time exposes the bay to the vacuum of space. Neither alarm alone means this. The PLC has no single output for the combination, so the operator must read both alarms for the same container.

8. OPERATOR PROCEDURES

8.1 ROUTINE CHECKS

1) Confirm the camera feed is live and all three cargo lights are on. Both should be active whenever the shutdown signal is inactive.
2) Confirm the pressure alarm is silent. This means door_1, door_2 and door_3 all read TRUE (closed).
3) Note any active container alarms. Each one means that container is missing. Confirm that the door of each affected container is closed: alarm 0 with door_1, alarm 1 with door_2, alarm 2 with door_3.
4) Confirm the status of the door you are about to use before handling cargo. Remember that door_0 (bay entry) is not watched by the pressure alarm.

8.2 IF A CONTAINER ALARM IS ACTIVE (CONTAINER MISSING)

1) Identify which alarm is active. Alarm n corresponds to sensor_n and container n, so alarm 0 means sensor_0 reads TRUE and container 0 is missing.
2) Check that the door of that container reads closed (TRUE): door_1 for container 0, door_2 for container 1, door_3 for container 2. With the door closed, the alarm is acceptable, the bay stays sealed, and the other containers are unaffected.
3) Keep that container door closed. If the door is open, the container is missing and its door is open: go to Section 8.4 immediately.
4) Report the condition per RASA procedure before resuming handling of that container.

8.3 IF THE PRESSURE ALARM IS ACTIVE (A CONTAINER DOOR IS OPEN)

1) Check door_1, door_2 and door_3. At least one reads FALSE (open), which is what turns the alarm on. FALSE on door_1 means the door of container 0 is open, door_2 means container 1, and door_3 means container 2.
2) For each open door, check the alarm of the same container (alarm 0 for door_1, alarm 1 for door_2, alarm 2 for door_3). If that alarm is ON, the container is missing: go to Section 8.4 immediately.
3) If that container's alarm is OFF, the container is present and the door opens onto it, so the bay is not exposed to the vacuum. Close the door by sending it a close command (door_n_command = FALSE) and confirm its status changes to TRUE.
4) The pressure alarm clears by itself once all three container doors read TRUE. Report the event per RASA procedure.

8.4 IF A CONTAINER IS MISSING AND ITS DOOR IS OPEN (EMERGENCY)

[WARNING] Both alarms are ON for the same container. The bay is exposed to the vacuum of space.

1) Close the open container door at once by sending it a close command (door_n_command = FALSE) and confirm its status changes to TRUE.
2) The pressure alarm clears once all three container doors read TRUE. The container alarm stays on until the container is back in place and its sensor reads FALSE.

8.5 IF THE CAMERA AND LIGHTS ARE OFF

Loss of both the camera feed and all three lights at the same time indicates that the shutdown signal is active (Section 7.5). Do not attempt to work in the bay unlit or unobserved; report the shutdown state to the responsible authority.

8.6 IF YOU HAVE ANY QUESTIONS

Review the SHIFT_BRIEF_<SHIFT_NUMBER> for your specific instructions for your shift, this can be found in the file archives on the uplink. If you need

information from the previous shift a voice log can be found by navigating to the /audio/SHIFT_LOG_<SHIFT_NUMBER> on the uplink, if the previous operator did their due diligence following procedure.

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